awslabs / generative-ai-cdk-constructs Public

Code | Issues 12 | Pull requests 8 | Discussions | Actions

main | 21 Branches | 409 Tags | Go to file | Go to file | Code

dependabot[bot] chore(deps): bump lodash from 4.17.23 to 4.18.1 (#1298) c9c16e6 · 2 weeks ago

.github	chore(gh): update workflow (#1195)	5 months ago
.husky	feat: add husky with a convential commit ...	2 years ago
.projen	feat(S3 vectors): add S3 vectors L2 const...	3 months ago
apidocs	chore(deps): upgrade list of models and D...	2 weeks ago
docs	feat(deprecation): remove lambda layer co...	2 years ago
lambda	fix(deps): update Python runtime to 3.13 (...	last month
layer	fix(bda): uopdated botocore for bda	last year
projenrc	chore(gh): update workflow (#1195)	5 months ago
rosetta	chore(rosetta): final documentation compi...	6 months ago
src	chore(deps): upgrade list of models and D...	2 weeks ago
test	fix(deps): update Python runtime to 3.13 (...	last month
.eslintrc.json	chore(project): upgrade dependencies (#1...	6 months ago
.gitattributes	chore(deps): upgrade dependencies (#12...	3 months ago
.gitignore	chore(project): upgrade dependencies (#1...	6 months ago
.mergify.yml	chore(project): upgrade dependencies (#1...	6 months ago
.node-version	feat!: public release (#126)	3 years ago
.npmignore	feat(bedrock): add Neptune/GraphRAG su...	last year
.ort.yml	feat(ci): Add repository configuration and ...	3 years ago
.projenrc.ts	feat(S3 vectors): add S3 vectors L2 const...	3 months ago
AGENTS.md	chore(doc): start ai readiness of the repo (...	3 months ago
CHANGELOG.md	feat(S3 vectors): add S3 vectors L2 const...	3 months ago
CLAUDE.md	chore(doc): start ai readiness of the repo (...	3 months ago
CODE_OF_CONDUCT.md	chore: initial documentation (#5)	3 years ago
CONTRIBUTING.md	chore(PR): doc update and code generati...	last year
DEPRECATION_PROCESS.md	chore(PR): doc update and code generati...	last year
DESIGN_GUIDELINES.md	chore(PR): doc update and code generati...	last year
DEVELOPER_GUIDE.md	chore(doc): start ai readiness of the repo (...	3 months ago
DOCUMENTATION_TESTING.md	chore(rosetta): final documentation compi...	6 months ago
LICENSE	chore: project created with projen (#3)	3 years ago
	feature(license): Add NOTICE, Remove TH...	3 years ago

NOTICE		
README.md	chore(doc): update go package location (...	3 months ago
commitlint.config.js	feat: add commitlint github action	3 years ago
header.js	feat: Feature/ingestion construct (#8)	3 years ago
package.json	chore(deps): upgrade dependencies (#12...	2 weeks ago
repolinter.json	fix: repolinter compliance (#596)	2 years ago
tsconfig.dev.json	chore(deps): upgrade dependencies (#10...	last year
yarn.lock	chore(deps): bump lodash from 4.17.23 to ...	2 weeks ago

README | Code of conduct | Contributing | Apache-2.0 license | Security

AWS Generative AI CDK Constructs

STABILITY EXPERIMENTAL

> All classes are under active development and subject to non-backward compatible changes or removal in any future version. These are not subject to the Semantic Versioning model. This means that while you may use them, you may need to update your source code when upgrading to a newer version of this package.

View on Construct Hub

pypi v0.1.316 | npm v0.1.316 | nuget v0.1.316 | maven-central v0.1.281 | go v0.1.316

bedrock module Terraform | opensearchserverless module Terraform | sagemaker endpoint module Terraform | streamlit module Terraform

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Introduction

The AWS Generative AI Constructs Library is an open-source extension of the AWS Cloud Development Kit (AWS CDK) that provides multi-service, well-architected patterns for quickly defining solutions in code to create predictable and repeatable infrastructure, called constructs. The goal of AWS Generative AI CDK Constructs is to help developers build generative AI solutions using pattern-based definitions for their architecture.

The patterns defined in AWS Generative AI CDK Constructs are high level, multi-service abstractions of AWS CDK constructs that have default configurations based on well-architected best practices. The library is organized into logical modules using object-oriented techniques to create each architectural pattern model.

CDK Versions

AWS Generative AI CDK Constructs and the AWS CDK are independent teams and have different release schedules. Each release of AWS Generative AI CDK Constructs is built against a specific version of the AWS CDK. The CHANGELOG.md file lists the CDK version associatec with each AWS Generative AI Constructs release. For instance, AWS Generative AI CDK Constructs v0.0.0 was built against AWS CDK v2.96.2. This means that to use AWS Generative AI CDK Constructs v0.0.0, your application must include AWS CDK v2.96.2 or later. You ca continue to use the latest AWS CDK versions and upgrade the your AWS Generative AI CDK Constructs version when new releases become available.

Contributing

Contributions of all kinds are welcome! Check out our contributor guide

Design guidelines and Development guide

If you want to add a new construct to the library, check out our design guidelines, then follow the development guide

Getting Started

- *TypeScript*
- *Python*
- *NuGet*
- *Go*
- *Java*

Refer to the documentation for additional guidance on a particular construct: Catalog

Catalog

The following constructs are available in the library:

L3 constructs

Construct	Description	AWS Services used
SageMaker model deployment (JumpStart)	Deploy a foundation model from Amazon SageMaker JumpStart to an Amazon SageMaker endpoint.	Amazon SageMaker
SageMaker model deployment (Hugging Face)	Deploy a foundation model from Hugging Face to an Amazon SageMaker endpoint.	Amazon SageMaker
SageMaker model deployment (Custom)	Deploy a foundation model from an S3 location to an Amazon SageMaker endpoint.	Amazon SageMaker
Amazon Bedrock Monitoring (Amazon CloudWatch Dashboard)	Amazon CloudWatch dashboard to monitor model usage from Amazon Bedrock.	Amazon CloudWatch
Bedrock Data Automation	Use Amazon bedrock data automation client to to build and manage intelligent document processing, media analysis, and other multimodal data-centric automation solutions	AWS Lambda, Amazon S3 bucket
Bedrock Batch Step Functions	Manage Bedrock model invocation jobs(batch inference) in AWS Step Functions state machines	AWS Step Functions, AWS Lambda, AWS EventBridge, Amazon Bedrock, AWS IAM

L2 Constructs

> ⚠ Warning
>
> Important: Amazon Bedrock L2 constructs are transitioning to the AWS CDK core repository. You can now find these constructs at: https://github.com/aws/aws-cdk/tree/main/packages/%40aws-cdk/aws-bedrock-alpha. Please migrate to the alpha package, as Bedrock L2 constructs in this repository are now deprecated and will no longer receive updates.

Construct	Description	AWS Services used
Amazon Bedrock	CDK L2 Constructs for Amazon Bedrock.	Amazon Bedrock, Amazon OpenSearch Serverless, AWS Lambda
Amazon OpenSearch Serverless Vector Collection	CDK L2 Constructs to create a vector collection.	Amazon OpenSearch Vector Index
Amazon OpenSearch Vector Index	CDK L1 Custom Resource to create a vector index.	Amazon OpenSearch Serverless, AWS Lambda
Amazon Aurora DSQL	CDK L2 Constructs for Amazon Aurora DSQL.	Amazon Aurora DSQL
Amazon S3 Vectors	CDK L2 Constructs for Amazon S3 vectors.	Amazon S3 Vectors

Sample Use Cases

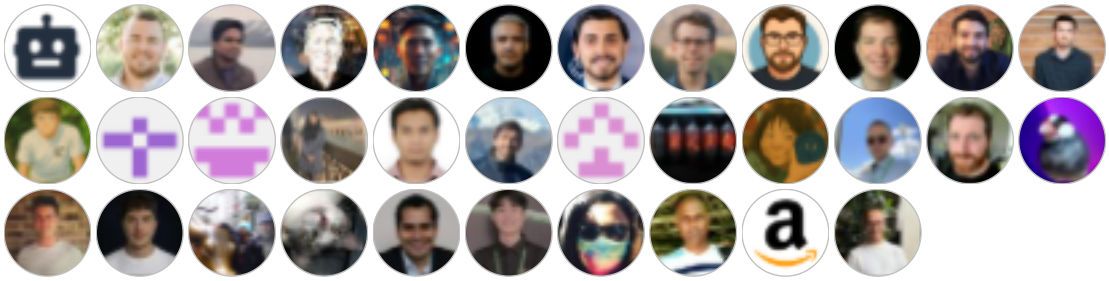
The official samples repository includes a collection of functional use case implementations to demonstrate the usage of AWS Generative A CDK Constructs. These can be used in the same way as architectural patterns, and can be conceptualized as an additional "higher-level" abstraction of those patterns. Those patterns (constructs) are composed together into stacks, forming a "CDK app".

Additional Resources

Resource	Type	Description
AWS re:Invent 2023 - Keynote with Dr. Werner Vogels	Keynote	Dr. Werner Vogels, Amazon.com's VP and CTO, announces the AWS Generative AI CDK Constructs during his AWS re:Invent 2023 keynote.
Workshop - Building Generative AI Apps on AWS with CDK	Workshop	In this workshop, you will explore how to build a sample generative AI app on AWS using CDK and Generative AI CDK Constructs.
Workshop - Hands on AWS CDK Generative AI Constructs	Workshop	In this workshop you will deploy projects that use CDK constructs from this library. Projects are from the amazon-bedrock-samples Github Repository.
Build generative AI applications with Amazon Titan Text Premier, Amazon Bedrock, and AWS CDK	Blog post + Code sample	Blog post exploring building and deploying two sample applications powered by Amazon Titan Text Premier using the Generative AI CDK constructs.
aws-cdk-stack-builder-tool	Code sample	AWS CDK Builder is a browser-based tool designed to streamline bootstrapping of Infrastructure as Code (IaC) projects using the AWS Cloud Development Kit (CDK).
CDK Live! Building generative AI applications and architectures leveraging AWS CDK Constructs!	Video	CDK Live! episode focused on building and deploying generative AI applications and architectures on AWS using the AWS Cloud Development Kit (CDK) and the AWS Generative AI CDK Constructs.
Announcing AWS Generative AI CDK Constructs!	Blog post	Blog post announcing the release of the AWS Generative AI CDK Constructs.
Streamline insurance underwriting with generative AI using Amazon Bedrock	Blog post + Code sample	Blog post and code sample discussing how to use AWS generative artificial intelligence (AI) solutions like Amazon Bedrock to improve the underwriting process, including rule validation, underwriting guidelines adherence, and decision justification.

Resource	Type	Description
aws-genai-llm-chatbot	Code sample	Multi-Model and Multi-RAG Powered Chatbot Using AWS CDK on AWS allowing you to experiment with a variety of Large Language Models and Multimodal Language Models, settings and prompts in your own AWS account.
bedrock-claude-chat	Code sample	AWS-native chatbot using Bedrock + Claude (+Mistral).
amazon-bedrock-rag	Code sample	Fully managed RAG solution using Knowledge Bases for Amazon Bedrock.
Amazon Bedrock Multimodal Search	Code sample	Multimodal product search app built using Amazon Titan Multimodal Embeddings model.
Amazon Bedrock Knowledge Bases with Private Data	Blog post + Code sample	Blog post and associated code sample demonstrating how to integrate Knowledge Bases into Amazon Bedrock to provide foundational models with contextual data from private data sources.
Automating tasks using Amazon Bedrock Agents and AI	Blog post + Code sample	Blog post and associated code sample demonstrating how to deploy an Amazon Bedrock Agent and a Knowledge Base through a hotel and spa use case.
Agents for Amazon Bedrock - Powertools for AWS Lambda (Python)	Code sample	Create Agents for Amazon Bedrock using event handlers and auto generation of OpenAPI schemas.
Text to SQL Bedrock Agent	Code sample	Harnessing the power of natural language processing, the "Text to SQL Bedrock Agent" facilitates the automatic transformation of natural language questions into executable SQL queries.
Dynamic Text-to-SQL for Enterprise Workloads with Amazon Bedrock Agent	Code sample	Elevate your data analysis with an end-to-end agentic Text-to-SQL solution, built on AWS for enterprise-scale adaptability and resilience. Ideal for complex scenarios like fraud detection in financial services.

Contributors



Operational Metrics Collection

Generative AI CDK Constructs may collect anonymous operational metrics, including: the region a construct is deployed, the name and version of the construct deployed, and related information. We may use the metrics to maintain, provide, develop, and improve the constructs and AWS services.

Roadmap

Roadmap is available through the GitHub Project

Deprecation

To understand our deprecation process, please refer to the dedicated documentation

License

Apache-2.0

Legal Disclaimer

You should consider doing your own independent assessment before using the content in this library for production purposes. This may

Releases 409

v0.1.316 Latest
on Mar 19

+ 408 releases

Packages

No packages published

Contributors 38

+ 24 contributors

Languages

TypeScript 92.4% Python 7.5% Other 0.1%

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